



Agentic Memory-Augmented Neural Machine Translation for English-Hindi with LoRA- Optimized Large Language Models

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Introduction

Seq-to-seq Models

Transformers

- Linear translation
- Static knowledge
- No self-correction
- Limited context
- Weak for low-resource

General Purpose

LLMs

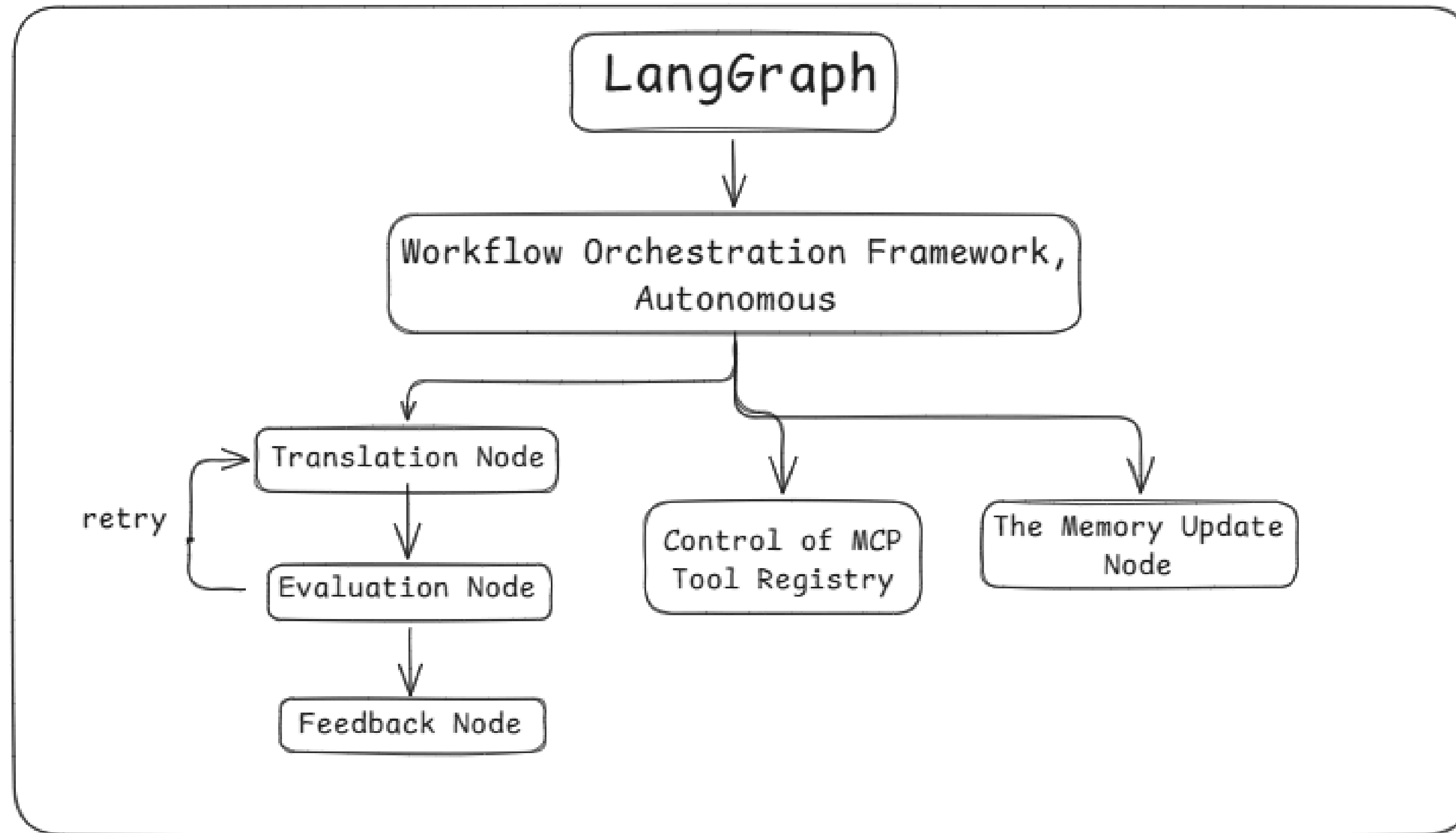
- Fluent translations
- Prompt-based adaptation
- Moderate reasoning
- No iteration loop
- Limited consistency

Agentic AI

Translation

- Iterative reasoning loop
- Self-correction via agents
- Memory-driven consistency
- Dynamic knowledge access
- Strong low-resource handling

Agentic AI Framework



Fine Tuning Process

Samanantar Dataset (10,000 pairs)



Preprocessing & Formatting



4-bit Quantization of OPT-1.3B



LoRA Adapter Injection



Forward Pass → Loss Computation



Backward Pass → Gradient Update (LoRA only)



Repeat for 3 epochs (30,000 steps total)



Save LoRA Adapter

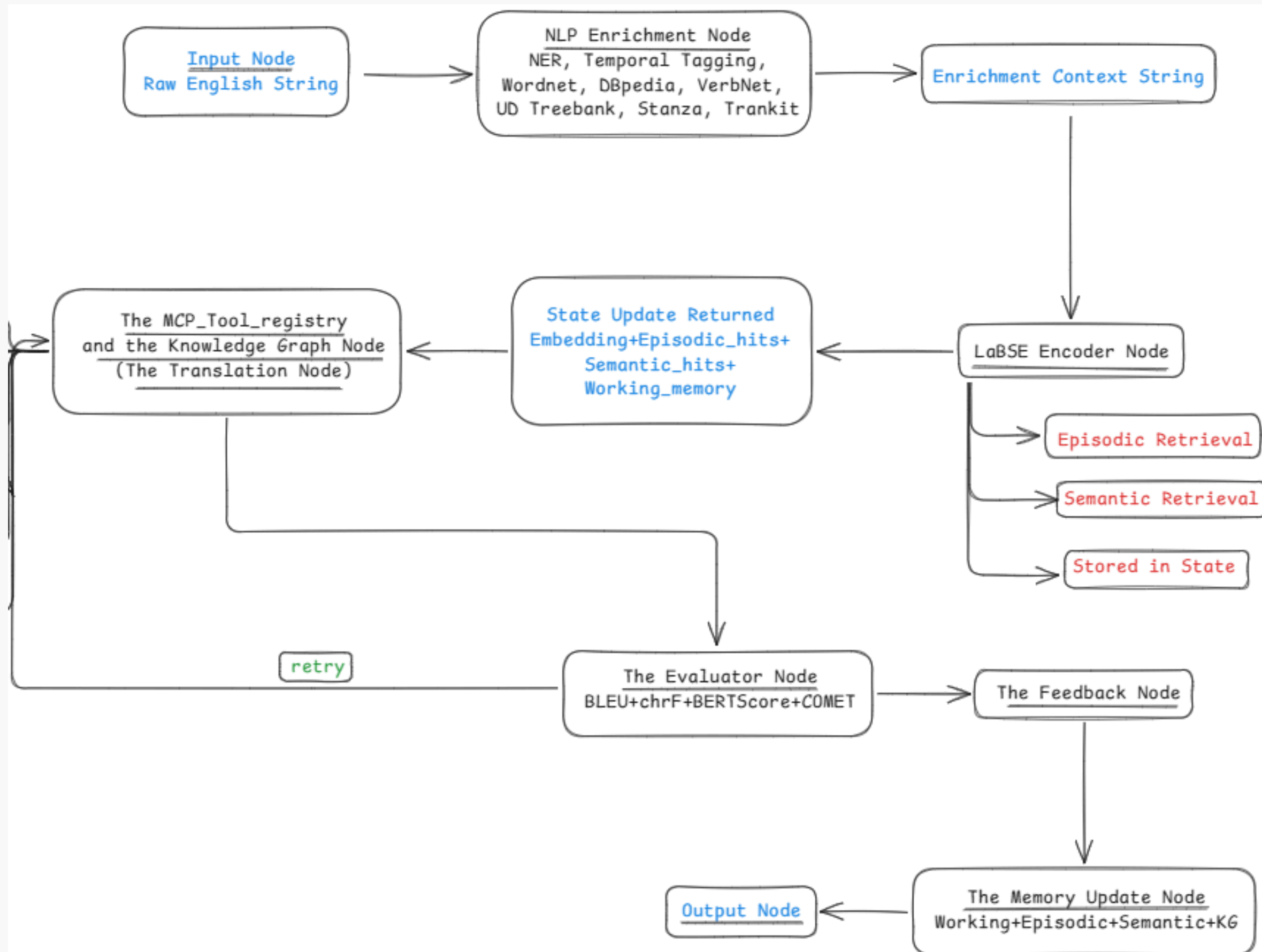


Fine-tuned Translation Model

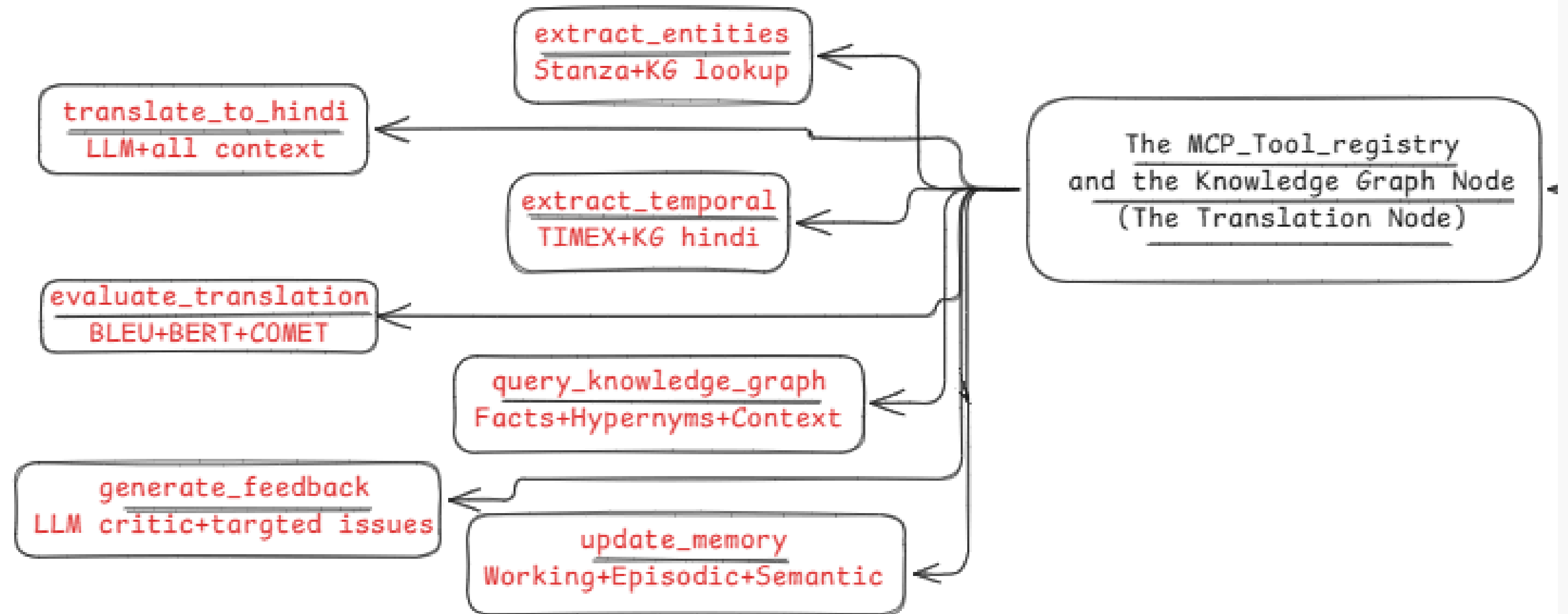
Objective of the Proposed Model

To design a memory-augmented, agentic machine translation system for English-to-Hindi that improves semantic accuracy, consistency, and robustness through iterative feedback, multi-agent collaboration, and low-resource adaptation.

Proposed Model



Proposed Model



Evaluation Metrics

BLEU (Bilingual Evaluation Understudy)

- Measures the lexical overlap between the machine's output and a human reference.
- Calculates n-gram precision.
- Fast and model-agnostic.

chrF (Character n-gram F-score)

- Uses character n-grams, and calculates an F-score (balancing precision and recall).
- More effective for morphologically rich languages.

Evaluation Metrics

BERTScore

- Uses a pre-trained model (BERT/RoBERTa) to create embeddings for the machine translation and the human reference. Then calculates the cosine similarity between these vectors.
- Rewards paraphrasing.

COMET (Cross-lingual Optimized Metric for Evaluation of Translation)

- Trained neural model, takes three inputs: the Source text, the Machine Translation, and the Human Reference.
- It is source-aware.

Results and Analysis

Performance Evaluation of the Proposed Agentic Machine Translation Model using a Large Language Model via Groq API

Metric	Mean Score
BLEU	34.48
chrF	64.56
BertScore	0.925
COMET	0.85

Results and Analysis

English to Hindi Translation Benchmark Table

Research/Task	Model	Dataset	Metric	Score
MDPI (2022)	Transformer	Samanantar	BLEU	33.45-33.72
Meta AI (2022)	NLLB-200	Flores-200	spBLEU	31.4
AI4Bharat (2023)	IndicTrans2	Flores-200	BLEU	35.8
AI4Bharat (2023)	IndicTrans2	IN22-Gen	chrF++	63-66

Results and Analysis

Proposed Model

Architecture: Facebook/opt-1.3b

Fine-tuning method: LoRA

Dataset: Ai4bharat/samanantar

Language Pair: En-Hi

Train Size: 10000

Epochs: 3

Dataset Analysis

Dataset	Split	Size	Model Training and Testing	Role
Samanantar	Train	~49M	10,000	Core training
FLORES-200	Test	~3K	200	Final Evaluation

Results and Analysis

Performance Evaluation of the Proposed Agentic Machine Translation Model using Fine-Tuning Method and testing on FLORES-200

Evaluation Metric	IndicTrans2_3.3B	mBART50	Proposed Model
BLEU	15.75	14.8	9.14
chrF	33.21	27.59	29.87
BERTScore	0.6142	0.5738	0.5983
COMET	0.582	0.5192	0.5267

Output

```
=====
Category : entity+technical
English  : ISRO launched a satellite from the Bengaluru facility this morning.
Reference: इसरो ने आज सुबह बेंगलुरु सुविधा से एक उपग्रह लॉन्च किया।
Output   : इसरो ने बेंगलुरु सुविधा से इस सुबह एक उपग्रह प्रक्षेपित किया।
Entities : [('ISRO', 'ORG', 'इसरो'), ('Bengaluru', 'GPE', 'बेंगलुरु'), ('this morning', 'TIME', '')]
Temporal : [('morning', 'सुबह')]
KG facts : 2
```

Iter	BLEU	chrF	BERTScore	COMET	Aggregate
1	21.83	59.65	0.9039	0.9425	0.6996

```
=====
[Node 1 – Input] 'Rajesh Gavre, the President of the MNPA teachers association, honoured...'
[Node 2 – Enrichment] entities=2 temporal=0 senses=7
[Node 3 – KG] facts=0 kg_context_len=0
[Node 4 – Embedding] dim=768 epi=1 sem=5
INFO:pytorch_lightning.utilities.rank_zero:GPU available: False, used: False
INFO:pytorch_lightning.utilities.rank_zero:TPU available: False, using: 0 TPU cores
INFO:pytorch_lightning.utilities.rank_zero: ; Tip: For seamless cloud logging and experiment tracking, try installing [litlogger](https://pypi.org/pr
INFO:pytorch_lightning.utilities.rank_zero: ; Tip: For seamless cloud uploads and versioning, try installing [litmodels](https://pypi.org/project/lit
/usr/local/lib/python3.12/dist-packages/pytorch_lightning/utilities/_pytree.py:21: `isinstance(treespec, LeafSpec)` is deprecated, use `isinstance(tr
[Node 5 – Translation] iter=1: 'राजेश गावड़े, एमएनपीए शिक्षक संघ के अध्यक्ष, ने पुरस्कार प्र...'
Predicting: 0it [00:00, ?it/s][Node 6 – Evaluator] PASSED agg=0.999 BLEU=0.0 BERT=0.999
[Node 7 – Feedback] Translation accepted. aggregate=0.999 BLEU=0.0 BERTScore=0.999
[Node 8 – Memory] episodic=118 semantic=151 kg_updated=True
[Node 9 – Output] 'राजेश गावड़े, एमएनपीए शिक्षक संघ के अध्यक्ष, ने पुरस्कार प्रदान करके विद्यालय को सम्मानित किया।'
```

Iter	BLEU	chrF	BERTScore	COMET	Aggregate
1	0.00	0.00	0.9985	0.9985	0.9985

Agent trace: input_node → enrichment_node → kg_node → embedding_node → translation_node_iter1 → evaluator_node → feedback_node → memory_update_node

```
=====
```


Model Analysis

1. The Enrichment Node

- Stanza NER uses a BiLSTM-CRF neural model trained on CoNLL data to label every token.
- TIMEX tagging uses regex patterns.
- WordNet sense disambiguation tokenises the sentence, POS-tags it with NLTK and then proceed with queries.
- DBpedia SPARQL fires a real HTTP query.

2. The Knowledge Graph Node

- Dynamic, for each entity that is recognised and translated in that session gets added via memory update node.
- Backed by NetworkX.

Model Analysis

3. The MCP Tool Registry

- Name-Value Python Dictionary.
- LangGraph node initiates the call for a particular entity, MCP client looks up the tool by name and invoking it with typed arguments.

4. The Embedding Node

- Encodes the sentence with LaBSE.

5. The Translation Node

- Semantic Knowledge from ChromaDB.
- Episodic Examples from past translations.
- Working Memory from current session.
- Enrichment context.
- KG context.

Model Analysis

6. The Evaluator Node

- BLEU, BERTScore, chRf, COMET
- Updates all three memory tiers.
- Add new entity pairs to the KG.
- Proceed either to the output node or retry the translation.

Future Scope and Conclusion

Future Work:

1. Support to a wider range of low-resource languages.
2. Dealing with the adversarial examples and improving adversarial robustness.
3. Optimizing the agentic framework for latency and deployment in real-time applications.

Conclusion:

This thesis presented a comprehensive approach to English-to-Hindi machine translation by integrating agentic artificial intelligence with memory-augmented neural architectures, enabling modular and autonomous decision-making model.

Referred Papers

- 1) Are AI agents the new machine translation frontier? Challenges and opportunities of single- and multi-agent systems for multilingual digital communication, **Journal: arXiv, Publication Year: 2025**
- 2) Agent-as-a-Graph: Knowledge Graph-Based Tool and Agent Retrieval for LLM Multi-Agent Systems, **Journal: arXiv, Publication Year: 2025**
- 3) Leveraging the Power of Large Language Models in Entity Linking via Adaptive Routing and Targeted Reasoning, **Journal: Association for Computational Linguistics, Publication Year: 2025**
- 4) Transformers: Leveraging OpenNMT and Transfer Learning for Low-Resource Indian Language Translation, **Journal: Association for Computational Linguistics, Publication Year: 2025**
- 5) Reinforcement of low-resource language translation with neural machine translation and backtranslation synergies, **Journal: International Journal of Artificial Intelligence, Publication Year: 2024**
- 6) mT5: A massively multilingual pre-trained text-to-text transformer, **Journal: arXiv, Publication Year: 2021**



Thank you

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